

Electric Potential Energy

Goal. Understand where electric potential energy comes from — summing over pairs of point charges, or pushing a charge against a field — and use it to find the potential energy of an electric dipole in a uniform field. Along the way: a system can have *negative* PE, and the configuration you call “zero” is not always the lowest one available.

1. Potential energy of two point charges (from the class prep)

Pulling a pair of point charges apart, or pushing them together, takes work — work that gets stored as potential energy.

$$U(q_1, q_2) =$$

- Keep the _____ of the charges — U can come out positive *or* negative.
- As $r_{12} \rightarrow \infty$, $U \rightarrow$ _____.
- For three or more charges, sum over every _____ (not every charge).

$$U_{\text{total}} =$$

Check. Two $+q$ charges are brought from far apart to a separation r . Is U positive or negative? What about a $+q$ and a $-q$?

2. Why “zero” isn’t always the lowest

We usually call the ground $h = 0$ and say a ball resting on it has zero gravitational PE. Now picture a hole in the ground, right under the ball.

Check. If the ball falls into the hole, is its PE above or below the zero you originally chose? What does that tell you about calling a PE “zero”?

- Opposite charges (+ and -): U is _____ at any finite separation, and $U \rightarrow 0$ only as $r \rightarrow \infty$. So $U = 0$ is actually the _____ value here — pulling the charges closer always finds a *lower* (more negative) PE, like digging the hole deeper.
- Like charges (same sign): U is _____ at any finite separation, and again $U \rightarrow 0$ as $r \rightarrow \infty$. Here $U = 0$ is the _____ value — you can’t push like charges to a lower PE than infinitely far apart.

3. Potential energy from pushing against a field

A uniform field $\vec{E}$ pushes a positive charge q with force qE . Push the charge a distance d the *opposite* way — straight against the field.

$\Delta U =$

- This is exactly the ball-and-hole idea: you do _____ work pushing against the field, and that work becomes stored PE.
- Let the charge go, and it moves *with* the field instead: now ΔU is _____ — the field does the work, and spends the stored PE.

Check. A positive charge is released and moves along $\vec{E}$. Does its PE increase or decrease? Where did the energy go?

4. The dipole in a field

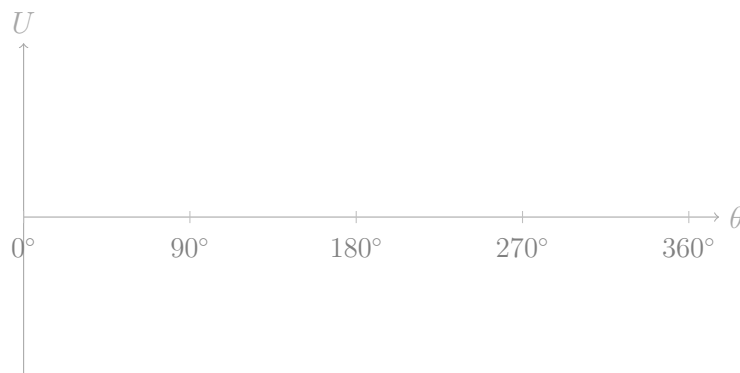
A dipole is a $+q$ and a $-q$ separated by $\vec{d}$ (dipole moment $\vec{p} = q\vec{d}$), sitting in a uniform field $\vec{E}$, tilted at angle θ between $\vec{p}$ and $\vec{E}$. Each charge sits at a different place along $\vec{E}$ — use §3's pushing idea on each charge, then add.

$$U(\theta) =$$

- At $\theta = 90^\circ$ (perpendicular to the field): $U =$ _____.
- At $\theta = 0^\circ$ (aligned with the field): $U =$ _____ — which is _____ than the $\theta = 90^\circ$ value!
- At $\theta = 180^\circ$ (anti-aligned): $U =$ _____.

Check. Which orientation is the most stable (lowest PE)? Is that the orientation where $U = 0$? What does this have in common with the ball and the hole?

5. Sketch it: U vs. θ



Check your curve, and the point-charge and uniform-field results above, against the *Assembling Charges*, *PE vs. Separation*, and *Electric Dipole in a Field* sims — open their **Info** panels.